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When venture capital comes to town: Local ecosystem effects beyond funded ventures☆

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ABSTRACT

When venture capital (VC) investment flows into a local economy, does it yield economic activity beyond the receiving ventures? While a large literature studies the effect of VC investments on venture-level activity, its consequences for broader economic activity are less well understood. We study the effect of VC investment flows on destination county employment, wages, and venture creation, using a novel instrument based on the 'social connectedness' of counties to major non-local sources of VC investment. Using detailed data on investor-to-venture VC flows, we find substantial positive effects: using instrumental-variable estimates, each million dollars invested into a county yields approximately one new venture, 41 jobs, and seven million dollars in payroll. We also document positive spillovers across industries, particularly toward those supplying VC-backed ventures—thereby contributing to the resource munificence of the local economy and its entrepreneurial ecosystem.

Executive summary

The venture capital industry has expanded dramatically since the dot-com bubble, with annual investment rising from $22 billion in 2004 to $241 billion in 2022. While a substantial literature documents the positive effects of VC on funded ventures—including employment growth, innovation, and productivity gains—the broader consequences for local economic activity remain less well understood. This paper examines whether VC investment generates economic activity beyond the funded ventures, contributing to local employment, venture creation, and the development of entrepreneurial ecosystems.

We study the effect of VC investment flows across U.S. counties on destination-county economic outcomes between 1986 and 2019, using detailed investor-to-venture data from ThomsonOne combined with County Business Patterns data from the U.S. Census Bureau. Our conservative estimates demonstrate that VC investment has substantial positive effects on local economic activity. Each million dollars invested produces at least one new venture, 41 jobs, and $7 million in payroll. In other words, a doubling of VC investment is associated with at least an 11% increase in ventures, an 8% increase in employment, and a 10% increase in payroll wages. These estimates capture combined direct effects on funded ventures and spillovers to other local firms. Decomposing these aggregates, we find that venture entry and job creation dominate exit and destruction, indicating that successful venture scaling outweighs portfolio failure rates.

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A core contribution concerns inter-industry spillovers, an aspect largely unexplored in prior work. We disaggregate outcomes to the industry-county-year level using 22 NAICS industries and distinguish between the effects of own-industry VC (which captures direct effects and competition between firms within the same industry) and other-industry VC (which captures spillovers across industries). We find that several industries benefit indirectly from VC investments they do not receive themselves. Notably, 'Management of Companies and Enterprises' (NAICS 55) and 'Professional, Scientific, and Technical Services' (NAICS 54) show the largest employment responses, primarily from other-industry VC. This finding is consistent with the expansion of firms that supply to VC-receiving ventures, in response to demand from VC-backed ventures. Importantly, these inter-industry effects accumulate over time: while initially smaller than within-industry effects, they converge over a five-year horizon. This pattern suggests that VC investment contributes to the resource munificence of local entrepreneurial ecosystems by expanding the supplier base that enables new ventures to access essential services.

Our results have several implications for policy and practice. First, the substantial local knock-on effects of VC investment suggest that policies that promote VC investment may generate broader economic benefits than previously recognized. Second, the role of social connectedness in attracting VC points to the potential value of initiatives that strengthen ties between local entrepreneurs and investors in established VC hubs—such as networking events, accelerator programs, and infrastructure investments that reduce travel time. Third, the evidence for supplier-industry spillovers suggests that VC's economic impact extends beyond high-tech sectors, potentially reducing entry barriers for future entrepreneurs by enhancing local business-service availability.

We provide the first systematic evidence of inter-industry spillovers from VC investment, demonstrating that the benefits extend beyond VC-receiving industries to upstream supplier sectors—thereby contributing to the development of local entrepreneurial ecosystems.

1. Introduction

When venture capital (VC) investment flows into a local economy, does it yield growth and entrepreneurial activity over and above the well-established positive effects for VC-receiving ventures? Venture capital (VC) investors differ from banks and other capital providers: they acquire detailed knowledge about portfolio companies and monitor them intensively. Consequently, "these inside investors can provide financing to young businesses that otherwise would not receive external funds" (Lerner, 1995, p.301). While VC clearly benefits recipient companies (e.g. Kortum and Lerner, 2000; Davila et al., 2003; Da Rin et al., 2013; Bernstein et al., 2016; Nanda et al., 2020), less is known about whether these investments translate into measurable effects on local economic activity beyond the funded company.

We examine this question for several reasons. First, the VC industry has expanded dramatically since the dot-com bubble, with annual investment rising from $22 billion in 2004 to $241 billion in 2022 (NVCA, 2023). This post-dot-com recovery marks a distinct period of VC expansion relative to earlier periods examined in the literature (e.g., Samila and Sorenson, 2011). Second, companies receiving VC are increasingly geographically dispersed beyond traditional tech hubs or investor locations, suggesting that VC may now contribute to economic activity more broadly across regions. Third, recent evidence indicates that VC investment tends to raise innovation in technologically related companies (Schnitzer and Watzinger, 2022), and that VC-backed startups contribute disproportionately to national innovation and growth (Akcigit et al., 2022).[1] Despite this evidence, skepticism persists as to whether VC can generate growth beyond its traditional high-tech strongholds, particularly given the increasing concentration of investment in certain technologies and sectors and the decline in corporate governance during the 2010s (Lerner and Nanda, 2020). Moreover, Guzman and Stern (2020) document a decline in entrepreneurial quality during the 2000s (a reduction in companies' ability to scale) that may or may not have reversed during the 2010s as venture capital investment increased and federal support for entrepreneurial clusters expanded (Chatterji et al., 2014).

In this paper we focus on the impact of VC on the local economy, including its entrepreneurial ecosystem, capturing both direct effects and cross-industry spillovers. First, we revisit whether the combined effects of VC-related innovation spillovers, weaker governance, and lower entrepreneurial quality in recent decades nonetheless generate local economic activity in terms of new ventures and additional jobs. We extend the analysis beyond VC investors' traditional home counties to capture the geographic decentralization of VC activity across U.S. counties and to capture the potential emergence of entrepreneurial ecosystems. Second, by disaggregating data by industry, we test whether the concentration of VC investment in a few high-tech sectors influences growth in other sectors—through innovation diffusion and input–output linkages—thereby contributing to the resource munificence of the local economy and its entrepreneurial ecosystem.

We collect detailed investor- and company-level data on the value of VC investments made between 1986 and 2019 and combine these with panel data from the US Census County Business Patterns on destination county total employment, total payrolls, and the count of establishments, as well as venture entry and exit, and job creation and destruction. We refer to an increase in the count of establishments (which is recorded by counting the 'physical locations at which business is conducted') as entrepreneurs seeing an opportunity to open a new business or venture in that location.[2]

To address endogeneity (e.g., reverse causality where VC investors target already-growing regions), we propose a novel instrument: 'Social Access to Venture Capital' (SAVC). The instrument rests on the premise that social connections facilitate

[1] VC also positively influences the rate of new business creation in 21 European countries (Popov and Roosenboom, 2013). Moreover, Popov (2014) document that U.S. industries receiving greater VC investment experience increases in average firm size, driven by a rising share of medium-sized and large firms.

[2] Although these are not necessarily innovative, growth-driven, or entirely new organizations. See Parker (2018) for a recent review on the definition and measurement of entrepreneurship.

investment: places with stronger social ties to VC investors receive more funding when those investors deploy capital. SAVC is defined as the sum of VC capital available in all source counties, weighted by the destination county's social ties to each source, and excludes all direct flows to the metropolitan area of interest.

We proxy social connections across states and counties using the Social Connectedness Index (SCI; Bailey et al., 2018), derived from Facebook friendship data. Using the same data, Kuchler et al. (2022) show that institutional investors are more likely to invest in publicly traded companies from regions to which they have stronger social ties, noting that "Facebook friendship links represent a broader type of network that is intrinsically different from networks based on factors such as shared neighborhood or education institutions" (p. 2780). Rehbein and Rother (2025) show that this relationship also holds for cross-county bank lending. By excluding VC flows into the destination county's own metropolitan area from the instrument, we mitigate simultaneity concerns (e.g., local agglomeration effects). Additionally, we include a rich set of fixed effects (including metropolitan area by year) to absorb other potential confounding factors.

Our results indicate that venture capital investment has a large and positive impact on local activity. In our preferred IV estimates, a doubling of VC investment is associated with at least a 4.7% increase in the number of business ventures, an 8.3% increase in jobs, and a 9.8% increase in total payroll. In dollar terms, each million dollars invested yields roughly one new venture, 41 jobs, and $7 million in payroll. These findings are robust to controlling for a broad set of fixed effects, various alternative definitions of the instrument, controls for overall market access and educational attainment of the local population, and the exclusion of major VC source states California, Massachusetts, and New York. Analysis of industry heterogeneity reveals that many industries benefit *indirectly*, likely by supplying to ventures that receive VC funding. We also find evidence of reallocation, in which industry growth comes at the cost of decline in other industries.

We provide a theoretical basis for the positive effects of local VC investment on the wider local economy by drawing on the literature on entrepreneurship and entrepreneurial ecosystem development. As we elaborate in the next section, the well-known direct effects on funded ventures contribute to local growth when the growth of successful investments outweighs portfolio failure rates—for example, when VC alleviates financial resource constraints and VC investors reduce environmental resource uncertainty and dependence, which helps to 'build winners' and enables firms to scale (Hillman et al., 2009). These effects may extend to the local economy through innovation spillovers, and when success stories transform latent entrepreneurial intent in others into action, contributing to the development of a local entrepreneurial ecosystem (Sarasvathy, 2008; Kerr and Coviello, 2020; Ebbers et al., 2026). Conversely, rapid scaling of successful ventures also competes for local resources (Folta et al., 2006), leading to reallocation from less productive to more productive companies, which mutes the *aggregate* impact of VC—even if such reallocation is economically efficient. Positive effects may also propagate, such as through inter-industry linkages and the growth of industries that supply to VC-backed ventures. This enables new ventures to access essential resources as 'producer services'-resources expand, and enhances the overall resource munificence of the local ecosystem (Stuart and Sorenson, 2003). In turn, this lowers entry barriers for other new ventures (Stam and van de Ven, 2021). Combined, these forces suggest that the benefits of VC investments are not confined to funded ventures and that VC investment has broader implications for local venture growth and local entrepreneurial ecosystems.

Our paper relates to a broad literature that studies the drivers of economic growth in regional clusters. We build on the seminal work of Samila and Sorenson (2011), who study the years 1993 to 2002 and focus on VC investment flows into the same county as the home county of the VC investor. They find robust positive effects on payroll wages and the total number of establishments, and positive but less robust effects on employment. We contribute by updating the sample period to include the years up to 2019, by expanding geographic coverage to all U.S. counties and proposing a new instrument for VC investments across all counties, but more importantly by splitting VC investment and outcomes by broad industry—to capture spillovers across industries. Related work on spatial clustering of economic activity and innovation includes Glaeser and Kerr (2009), Glaeser et al. (2010b,a, 2015), Carlino and Kerr (2015), Tareque et al. (2024), but does not focus specifically on the role of venture capital for local economic activity. Chen and Ewens (2025) find positive effects of VC on the geographic clustering of high-growth startups, but do not study the impact on the wider local economy.

Another literature studies alternative innovations in entrepreneurial finance, such as crowdfunding (Agrawal et al., 2015; Sorenson et al., 2016; Lambert et al., 2025), but venture capital remains an important source of risk capital for local economies and for funding innovation by startups (Kerr and Nanda, 2015; Howell et al., 2023). Furthermore, by studying the real economic impact of VC funding on the local economy, we also contribute to the literature that studies booms and busts in VC (Townsend, 2015; Bernstein et al., 2019; Janeway et al., 2021).

Finally, we contribute to the fast-growing literature on the role of social networks in finance, including recent studies leveraging Facebook's Social Connectedness Index. For example, Kuchler et al. (2022) and Rehbein and Rother (2025) examine the geographic spread of investments by institutional investors, and cross-county lending, respectively. Our study is the first to use the SCI to construct an instrument for local VC investment.

The rest of the paper is organized as follows. Section 2 develops the theoretical framework and hypotheses. Section 3 describes the data and presents descriptive statistics. Section 4 outlines the empirical strategy, Section 5 presents the results, and Section 6 analyzes industry-level spillovers. Section 7 discusses the implications of our findings and concludes.

2. Theoretical framework

We examine how venture capital investment can influence county-level economic outcomes, both through its direct effects on recipient firms and through spillovers that affect the broader industrial landscape. The spillover channel is particularly important

because VC is neither universally accessible nor consistently effective as a financing model across all technologies and industries, despite its notable success in certain high-tech sectors such as information technology (Lerner and Nanda, 2020). This raises the question of whether the benefits of VC investment nonetheless extend to other industries via spillover effects.

Venture capital is a central component of entrepreneurial ecosystems, yet its aggregate impact beyond the funded firm depends on the balance between successful ventures and failures, as well as on the extent to which spillovers from VC investment diffuse through local markets. We conceptualize VC investment not merely as firm-level financing that alleviates resource constraints, but as a localized positive shock that generates: (1) direct scaling effects within funded firms; (2) intra-industry spillovers that may stimulate entrepreneurial entry and innovation in other firms, while also intensifying competition for scarce resources; and (3) inter-industry spillovers to upstream and downstream industries, which can ultimately enhance a region's resource munificence. This framework integrates the focus on individual firm performance (e.g., Davila et al., 2003; Da Rin et al., 2013; Bernstein et al., 2016; Nanda et al., 2020) with the broader literature on agglomeration economies and clusters (e.g., Glaeser et al., 2010a; Duranton and Puga, 2020), examining how VC contributes to the local economy.

Direct effects on the receiving company. First, as is well established, VC serves as a catalyst for new ventures that typically find themselves constrained in their ability to acquire financial, physical and human capital, such that VC investment enables new ventures to grow. VC relieves financial and credit constraints, and VC investors that take a seat on the board of directors to monitor the venture also provide access to resources through their networks, reducing environmental resource uncertainty and dependence (Hillman et al., 2009). VC can also signal quality and boost reputation to overcome the liability of newness (Zott and Huy, 2007; Hallen et al., 2014) and reduce asymmetric information in the market for initial public offerings (Chemmanur et al., 2021). A large literature confirms that VC generates positive average treatment effects in receiving firms, on employment (Engel and Keilbach, 2007; Ando et al., 2025), on innovation, and on labor costs (to attract talent) (Bronzini et al., 2020). The evidence thus points to an effect over and above 'picking winners' (startups that would also grow fast without VC backing), and that the deep involvement of VC investors rather helps to 'build winners' (which may have larger aggregate effects). Given that VC targets firms with high scaling potential, these direct effects of successful ventures may be large relative to the local economy. On the other hand, the inherent riskiness of startups means that many ventures will fail, and the reduced emphasis on governance during the 2010s (Lerner and Nanda, 2020) in combination with lower entrepreneurial quality during the 2000s (Guzman and Stern, 2020) may have reduced some of the potential growth benefits of VC. However, the skewed distribution of VC returns suggests that successful "home runs" likely outweigh failures in the aggregate. The more ventures receive VC investment and the faster and larger they grow, the larger the effect on the local economy:

H1: VC investment increases total county-level employment and venture creation, as direct growth effects dominate portfolio failure rates.

Intra-industry dynamics: Knowledge spillovers versus competition. Second, within the specific industry receiving the investment, VC triggers two opposing forces: knowledge spillovers and competition for resources. First, innovation diffusion may generate positive externalities. The innovation channel—supported by recent evidence on patent spillovers (Schnitzer and Watzinger, 2022)—suggests that VC-backed firms attract further entry by providing knowledge capital (Keuschnigg, 2004). Moreover, the local apparent success of some entrepreneurs in attracting VC investment may also transform existing but latent entrepreneurial intent in others into action, creating more startups and spin-offs. This adds to the local ecosystem, expanding networks, that—through a process of effectuation and co-creation—may lead to further entry of new firms (Sarasvathy, 2008; Kerr and Coviello, 2020; Ebbers et al., 2026). Second, however, the entry of startups and the rapidly scaling of firms competes for local resources (Folta et al., 2006), particularly skilled and high-quality labor, potentially driving up wages and crowding out less productive incumbent firms (as in "creative destruction" models; Aghion and Howitt, 1992). A process of reallocation of physical and human capital that fuels startups may then lead to the decline of other less productive firms within the industry because they likely use similar physical and human capital, attenuating the *aggregate* impact of VC investment on employment and the number of ventures. On the other hand, reallocation of labor may be good for workers because competition for workers raises the total county payroll when they switch to better paying jobs. Which of the two forces of spillovers and competition dominates is an empirical question, but we expect:

H2a: VC investment increases average payroll wages in the local county due to labor market competition and productivity gains;

H2b: VC investment increases real outcomes (employment and new ventures) in the receiving industry in a county.

Evidence against H2b would suggest that the local competition effect is strong such that more firms close and layoff workers than can be absorbed by new startups, and/or that innovation does not spread to other firms.

Inter-industry linkages and entrepreneurial ecosystem development. Third, effects of VC investments propagate to other industries in the ecosystem via input–output linkages (Glaeser and Kerr, 2009; Glaeser et al., 2015). Local upstream suppliers (e.g. legal and notary services, back-office services, real estate, engineering, computing, advertising, consultancies, etc.) will benefit from customers that grow fast. Downstream industries that buy from (business-to-business) ventures also benefit if product innovation leads to cost reductions or quality improvements. Unlike intra-industry peers, these diverse sectors are less likely to compete for the exact same specialized labor pool, such that positive spillovers likely dominate competition for resources. Inter-industry connections further strengthen the entrepreneurial ecosystem, as locally available suppliers enable new ventures to access essential resources (Stuart and Sorenson, 2003; Acs et al., 2017). While not being the only essential endowment of a productive ecosystem, the growth of suppliers does enhance the overall resource munificence of the region, expands the 'producer services'-resource provided by ecosystems (Stam, 2015), increases the potential supporter network of entrepreneurs (Newbert et al., 2013), and thereby lowers entry barriers (Stam and van de Ven, 2021). For example, upstream business services allow VC-backed and other startups to outsource these functions,

reducing their fixed costs. This enriched environment supports venture growth (Spigel and Harrison, 2018) and has the potential to set in motion a self-reinforcing cycle that accelerates entrepreneurial activity (Sorenson, 2017). However, relatively little is known about the impact of VC investment on the local economy of industries other than on the receiving industry itself, and whether there indeed is a positive aggregate effect on suppliers. We expect that:

H3: VC investment positively impacts employment and venture creation in non-receiving industries in the county (spillovers), particularly in upstream sectors.

In terms of timing, we expect a positive impact of direct effects to happen relatively soon after investment (H1 and H2a/b), which may be sustained over time due to positive spillovers. Medium-run effects depend on whether successful scaling of new ventures (and spillovers) dominate competition effects and the high failure rates of ventures (H2b). Inter-industry effects and any positive feedback from ecosystem development will likely take longer to materialize (H3).

Finally, while we focus on the county as the relevant economic unit, we acknowledge that spillovers may extend across space, since counties are connected through linkages like trade and movements of labor. Any effect of VC investment on county-level employment is mediated by the elasticity of labor supply (depending e.g. on the costs of migration, see Kleinman et al. (2023) for a formal treatment). We assume here that cross-county labor mobility is neither zero nor perfectly flexible. We further note that the costs associated with distance likely attenuate the geographic spillovers, such that our local approach captures the strongest effects. Finally, employment effects will also be more muted if (new) companies hire workers from the pool of unemployed in their own or from other counties.

3. Data

The dependent variables are taken from the County Business Patterns (CBP), published annually by the United States Census Bureau. In particular, we use annual observations of total employment, annual payroll wages, and the number of ventures, for all US counties between 1986 and 2019. *Employment* is the total of full- and part-time employees on the venture's payroll; *Payroll* includes all forms of compensation paid during the reporting year in thousands of dollars; *Ventures* is the count of individual physical locations at which business is conducted ('establishments').[3] We observe the total amount of businesses in each year, such that we can track venture creation *net* of closures over time. The final sample contains a balanced sample of 3125 counties, observed annually between 1986 and 2019.

The main explanatory variable is the total amount of venture capital (VC) equity investment into all companies in a given US county in a given year. This variable is constructed from data on all recorded individual venture capital investments at the firm level in the ThomsonOne database, which is one of the standard sources on VC investments used in the literature (Da Rin et al., 2013). We observe 240,978 investments by 6010 investors into 41,487 companies within the lower 48 states of the US. Using this investment level information, we geocode the reported addresses of the investor (a venture capital firm) and the receiving company. We then aggregate the dollar amounts invested into companies located in each destination county in each year.[4] Let the total amount of venture capital equity investment (reported in current millions of US dollars) into destination county i in year t be $VC_{i,t}$. Furthermore, let $VC_{ij,t}$ denote the bilateral VC investment flow into destination county i from origin county j in year t. We thus have that

$$VC_{i,t} = \sum_{j \in J} VC_{ij,t}$$

The Social Connectedness Index (SCI), a measure of social connections across US counties, was developed in collaboration with Facebook and published by Bailey et al. (2018). The index is based on Facebook friendships and measures the number of Facebook friendships between two counties divided by the product of the Facebook populations in the two counties. However, data on the county-to-county SCI is only available for a snapshot in August 2020. Previous studies (Kuchler et al., 2022; Wache, 2023) have shown that the influence of the SCI on several other variables is very stable over time, indicating that the county-level SCI is a slow-moving object, and that observations from 2020 are a reasonable proxy for aggregate county-level social connections in previous years. Denote the social connectedness index between county i and county j as SCI_{ij}. The index is defined as:

$$SCI_{ij} = \frac{\text{Facebook Connections}_{i,j}}{\text{Facebook Users}_i * \text{Facebook Users}_j} \tag{1}$$

The measure of the Social Connectedness Index that is publicly reported is then scaled between a maximum of 1,000,000,000 and a minimum of 1. A doubling of the SCI means that the chance that two random people are friends (one from each given location) is roughly twice as large.

According to Bailey et al. (2018), more than 58% of US adults and 71% of the US online population used Facebook as of September 2014. Although Facebook usage is relatively similar across income, education, and racial groups, it declines with age. The authors also report that Facebook is primarily used as a platform for real-world friends and acquaintances to interact online, and that people typically only add connections they know in real life. Based on these findings, the authors argue that Facebook data provides a unique representation of US friendship networks on a large scale.

[3] Source: https://www.census.gov/programs-surveys/cbp/technical-documentation/methodology.html

[4] We observe only one address of the receiving company: since they tend to be small and young companies, we assume that companies do not have other business locations outside of their headquarters' county.

Table 1
Descriptive statistics for county-level sample.

	N	Coverage	Mean	Median	SD	Min	Max
Dependent Variables							
Employment	105,753	1986–2019	34,702.0	6152	130,539.1	3	4,007,163
log(Employment)	105,753	1986–2019	8.8	8.7	1.7	1.1	15.2
Annual Payroll ($1000)	105,885	1986–2019	1,289,532.2	148,506	6,441,980.4	31	275,585,585
log(Annual Payroll)	105,885	1986–2019	12.0	11.9	1.9	3.4	19.4
Ventures	106,250	1986–2019	2254.1	532	7707.8	2	285,383
log(Ventures)	106,250	1986–2019	6.4	6.3	1.5	0.7	12.6
Venture Entry	104,400	1986–2019	226.2	48	823.6	0	29 919
log(Venture Entry)	104,081	1986–2019	4.1	3.9	1.5	1.1	10.3
Venture Exit	104,480	1986–2019	205.6	46	752.7	0	27 794
log(Venture Exit)	104,150	1986–2019	4.0	3.8	1.4	1.1	10.2
Job Creation	105,935	1986–2019	5141.4	832	20 568.2	0	727 103
log(Job Creation)	105,896	1986–2019	6.8	6.7	1.8	0	13.5
Job Destruction	105,935	1986–2019	4707.2	773	19 285.0	0	756 643
log(Job Destruction)	105,894	1986–2019	6.7	6.7	1.7	0	13.5
Explanatory Variables							
VC Equity Investment ($ mln)	106,250	1986–2019	5.7	0.0	103.6	0	11,570.3
asinh(VC Equity Investment)	106,250	1986–2019	0.2	0.0	0.9	0	10.0
Social Access to VC (SAVC)	106,237	1986–2019	.46	.24	1	.007	59.1
Social Access to GDP (SAGDP)	59,372	2001–2019	3.1	2.9	1	1.1	16.5
Education	12,082	2005–2019	27.6	26.1	10.2	7.3	76.2

Note: This table shows descriptive statistics for the main sample used to estimate Eq. (2) in Section 5. The unit of observation is a county-year.

Data on GDP at the county level (available from 2001) is taken from the Bureau of Economic Analysis. Information on the geography of the United States, such as Metropolitan Statistical Areas (MSA), the larger Combined Statistical Areas (CSA), and state membership at the county-level are taken from the website of the Census Bureau.[5] Descriptive statistics of all variables are provided in Table 1.

Within our sample, VC investments were made by 6010 individual investors: the top three source states are California (1610 investors), New York State (1115 investors), and Massachusetts (457 investors), reflecting that the VC industry is more clustered in space than the finance industry and entrepreneurship itself (Chen et al., 2010; Kraemer-Eis et al., 2018). However, our detailed data allows us to split origins and destinations: of the 240,978 observed investment flows, only 104,751 were done within the same state (39% of total value invested), and only 28,146 within the same county (10% of total value invested). In Fig. 1 we show the geographical distribution of VC investment by *destination* county, averaged over 2016–2019. This shows that inward investment is especially large in the Seattle, San Francisco, Los Angeles, Manhattan, and Boston areas. However, many other counties across the country receive VC investment: those with positive VC investment each receives on average 63.2 million dollars per year. This geographic pattern contrasts the higher geographical concentration of origin counties and motivates us to look at the effects of VC investment on destination counties.[6]

4. Empirical strategy

We develop the empirical strategy in four steps: Section 4.1 presents the baseline specification. Section 4.2 introduces our instrument—Social Access to Venture Capital (SAVC)—and Section 4.3 discusses identification. Section 4.4 extends the analysis to industry-level spillovers.

4.1. Baseline specification

We measure county-level economic performance using three main outcomes: total employment, total annual payroll, and the count of business ventures. We relate these to the total dollar amount of inward VC investment in the previous year. The baseline specification is:

$$Y_{i,t+1} = \beta VC_{i,t} + \gamma' X_{i,t} + \mu_i + \nu_{rt} + \varepsilon_{i,t}, \tag{2}$$

where $Y_{i,t+1}$ is the one-year-ahead value of annual payroll, employment, or ventures. $VC_{i,t}$ is the total VC equity investment into county i in year t. $X_{i,t}$ denotes control variables, μ_i denotes county fixed effects, and ν_{rt} captures region-by-year fixed effects where

[5] The US Census Bureau defines an MSA as "areas [that] consist of the county or counties (or equivalent entities) associated with at least one urbanized area of at least 50,000 population, plus adjacent counties having a high degree of social and economic integration with the core as measured through commuting ties". A CSA is defined as a group of adjacent metropolitan and micropolitan areas that has a high degree of interchange. See https://www.census.gov/programs-surveys/metro-micro/about/glossary.html. As of March 2020, there were 384 Metropolitan and 543 Micropolitan Statistical Areas, which together form 172 Combined Statistical Areas.

[6] In Figure OA1 we show the total amount of outward VC investment by *origin* county.

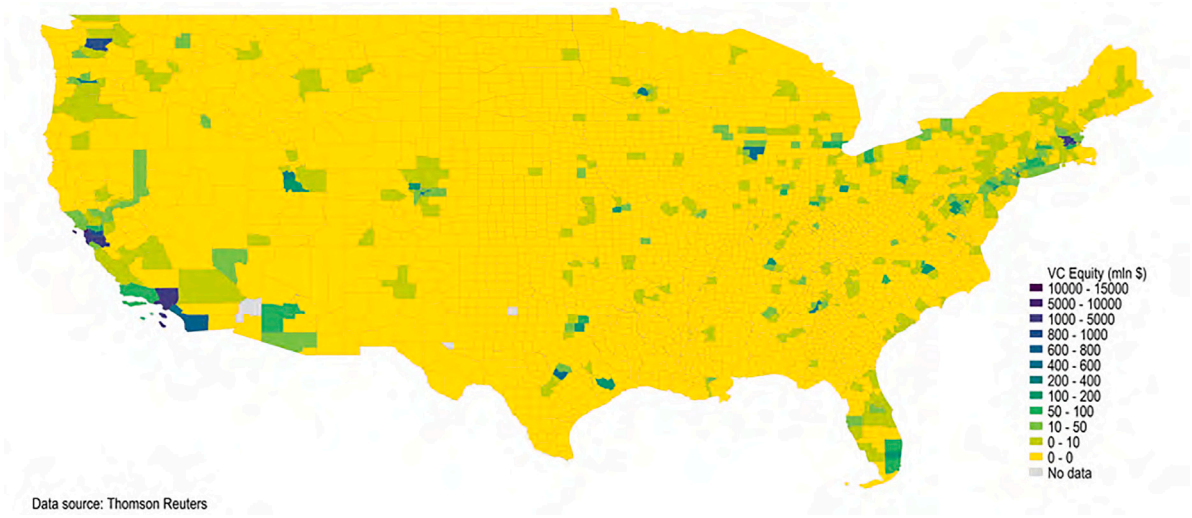

Fig. 1. Inward VC investments by destination county, average of 2016–2019.

r is alternatively the state (or CSA) r of county i. In most specifications we apply the inverse hyperbolic sine (asinh) transformation to $VC_{i,t}$ to accommodate zeros and skewness. Standard errors are clustered at the county level throughout, which accounts for heteroskedasticity and serial correlation.[7]

Although outcomes are measured one year after investment, estimating Eq. (2) by OLS raises endogeneity concerns. Counties with high growth may attract more VC, generating reverse causality; alternatively, local growth and VC investment may share common shocks. We address these concerns in two ways: first, by including a rich set of fixed effects (county, state-by-year, or CSA-by-year); second, by instrumenting $VC_{i,t}$.

4.2. Instrument: Social access to venture capital (SAVC)

Our proposed instrument, Social Access to Venture Capital (SAVC), captures a destination county's exposure to the nationwide supply of VC, weighted by social proximity to VC sources. The idea is to capture the likelihood that an entrepreneur in a given location is matched with a potential venture capital investor (from another location) to support the entrepreneur's business. Unlike banks, VC firms lack wide branch networks; matching between entrepreneurs and investors therefore depends heavily on social connections, especially for non-local deals.[8] In this spirit, we approximate the potential success rate of this unobserved matching process by a county's social proximity to venture capital, using the Social Connectedness Index (SCI) of Bailey et al. (2018), derived from Facebook friendships.

Formally, SAVC for county i in year t is:

$$SAVC_{i,t} = \sum_{j \in J} SCI_{ij} \cdot \sum_{k \notin M_i} VC_{jk,t} \tag{3}$$

where $SCI_{i,j}$ is the Social Connectedness Index between counties i and j, $VC_{jk,t}$ is the VC investment flowing from origin county j to destination county k in year t, and M_i is the set of counties in the same Metropolitan Statistical Area (MSA) as county i. Crucially, summing over all $k \notin M_i$ ensures that we exclude all flows into i's own MSA when constructing SAVC; this reduces mechanical correlation between the instrument and local outcomes.[9]

SAVC varies across counties through SCI_{ij} and over both time and counties through annual fluctuations in $VC_{jk,t}$. This captures one specific channel through which venture capital flows to a location: via the social connections to VC source counties. The construction parallels "market access" measures in trade (Harris, 1954; Redding, 2022), substituting social proximity for geographic proximity.

Fig. 2 illustrates the spatial pattern of SCI for San Mateo County (home to Menlo Park, a major VC cluster). Social ties decline with distance but not monotonically; beyond Colorado, connections follow urban centers. This non-linear geography generates variation in SAVC that is distinct from simple distance to VC hubs. To illustrate: Washoe County (Reno, Nevada) has no local VC firms, but strong social ties to California. When VC investors in San Mateo increase their activity, entrepreneurs in Washoe County—connected via social networks—become more likely to receive funding. SAVC aims to capture this channel.

[7] Online Appendix OA3 shows robustness to alternative clustering choices.

[8] Wache (2023) shows that the Social Connectedness Index is the most robust predictor of county-to-county VC flows, dominating physical distance, trade, migration, and commuting. See also Gompers et al. (2020) on how VCs source deals, and Kuchler et al. (2022) on how institutional investors rely on social ties.

[9] Alternative definitions of SAVC (e.g. excluding the own CSA or own county only) are examined in Online Appendix OA2; results are robust.

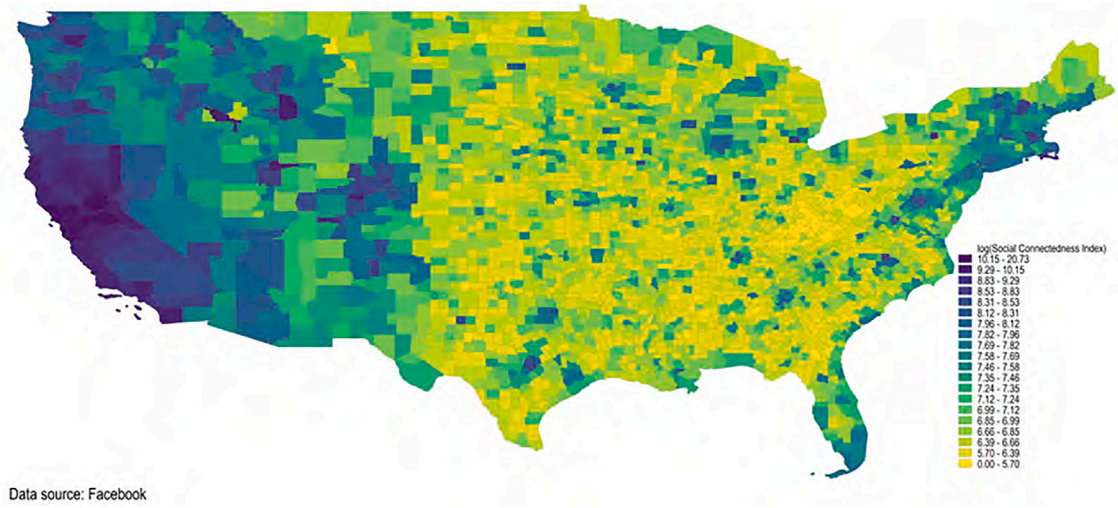

Fig. 2. SCI of San Mateo County, CA, (FIPS code: 06081), August 2020.
Note: This figure plots the SCI of San Mateo County (06081) with respect to all other counties in August 2020. The figure shows twenty quantiles (ventiles) of the distribution of log SCI.

4.3. Identification

The exclusion restriction requires that SAVC affects county-level outcomes *only* through its effect on local inward VC investments—i.e. conditional on controls, $SAVC_{i,t}$ is uncorrelated with $\epsilon_{i,t}$. We support this assumption with three strategies.

First, as stated earlier, we exclude all VC flows into the destination county's own MSA when constructing SAVC, which prevents the instrument from capturing direct mechanical effects of investments into the receiving county, or direct spillovers within the local labor market. Second, we include county fixed effects (absorbing time-invariant county characteristics) and either state-by-year or CSA-by-year fixed effects (absorbing time-varying regional shocks, changes in industrial composition, and local business cycles). This ensures identification comes from within-county over time variation, net of region-by-year shocks. Third, we control for general "market access" by including dummies for each percentile of $SAGDP_{i,t}$, defined as:

$$SAGDP_{i,t} = \sum_{j \neq i} SCI_{ij} \cdot GDP_{j,t} \tag{4}$$

where $GDP_{j,t}$ is county j's real GDP. In the baseline, we control for SAGDP by including a set of dummy variables for each percentile of SAGDP. These fixed effects absorb any benefits of social proximity to economic activity more broadly (Betz et al., 2018), ensuring that SAVC captures exposure to VC supply specifically rather than general connectedness to prosperous regions.

In addition, we show that results are robust to excluding the dominant VC source states of California, Massachusetts, and New York state from the sample (Online Appendix OA5).

4.4. Industry-level spillovers

The aggregate specification captures the combined direct effects and spillovers of VC investment. To separate these channels (H2b and H3), we disaggregate to the industry-county-year level.

This approach allows us to (i) identify which industries benefit most from own-industry VC, (ii) detect labor reallocation across industries, and (iii) test whether the effects of VC in one industry spills over into other industries.

Using data from Eckert et al. (2020), which harmonizes CBP employment to NAICS 2012 codes, we estimate for 22 two-digit top-level industries:

$$Y_{is,t+1} = \delta_1 VC_{is,t}^{Own} + \delta_2 VC_{is,t}^{Other} + \mu_{it} + \varphi_{st} + \nu_{rst} + \eta_{is,t}, \tag{5}$$

where $VC_{is,t}^{Own}$ is VC investment into industry s in county i in year t, and $VC_{is,t}^{Other} \equiv \sum_{s' \neq s} VC_{is',t}$ is VC investment into all other industries in the same county-year. μ_{it} denotes county-by-year fixed effects, φ_{st} denotes industry-by-year fixed effects, and ν_{rst} denotes industry-by-CSA-by-year fixed effects (included in some specifications). Standard errors remain clustered by county.

5. Impact of VC investment on local economic outcomes

How large is the impact of VC investments on local economic outcomes? This section quantifies the impact of VC investment on county-level employment, payroll, and ventures. OLS estimates are presented first (Section 5.1), which document positive

Table 2
OLS regressions of county-level aggregate outcomes on VC investments.

Dependent variable →	log Employment		log Payroll		log Ventures	
	(1)	(2)	(3)	(4)	(5)	(6)
Inward VC Investment, asinh, t-1	0.011***	0.005*	0.019***	0.012***	0.010***	0.004*
	(0.002)	(0.003)	(0.003)	(0.003)	(0.002)	(0.002)
Observations	102,604	102,637	102,736	102,769	103,092	103,125
Adjusted R^2	0.990	0.990	0.988	0.987	0.993	0.993
County FEs	Yes	Yes	Yes	Yes	Yes	Yes
State×Year FEs	Yes	No	Yes	No	Yes	No
CSA×Year FEs	No	Yes	No	Yes	No	Yes

Note: Standard errors clustered at the county level in parentheses. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$. *Employment* is the total of full- and part-time employees on the venture's payroll; *Payroll* includes all forms of compensation paid during the reporting year in thousands of dollars; *Ventures* is the count of individual physical locations at which business is conducted. *Inward VC Investment* is the inverse hyperbolic sine (asinh) of all inward VC equity investments in millions of dollars. CSA×Year FEs are combined statistical area-by-year fixed effects.

associations consistent with H1 (net positive direct effects) and H2a (rising payroll wages). We then address endogeneity through our SCI-based instrument (Section 5.2), summarize robustness checks (Section 5.3), and decompose effects into entry and exit (Section 5.4, testing H2b).

5.1. OLS results

We begin by establishing baseline correlations using OLS. For each of the outcomes—employment, payroll, and ventures—we show in Table 2 two specifications where we vary the fixed effects. Column 1 shows that, conditional on county and state-by-year fixed effects (controlling for permanent county characteristics and state-level business cycles), a doubling of inward VC investment is associated with an increase in employment by 1.1% in the following year. It is also associated with an increase in total payroll by 1.9% (column 3), and an increase in the number of ventures by 1.0% (column 5). Because the local business cycle may deviate from that of the state—potentially confounding both inward VC investments and local outcomes—we replace the state-by-year fixed effects with CSA-by-year fixed effects in columns 2, 4, and 6. The point estimates remain positive and significant, though attenuated, indicating that within-CSA variation still identifies sizable effects. Overall, these results are consistent with H1 (VC's direct growth effects dominate portfolio failures) and H2a (payroll rises, reflecting labor-market competition and productivity gains).

To investigate the timing of effects, we follow Jorda (2005) and Jorda and Taylor (2025) and project the impact of VC investment forward over five years. The annual point estimates β_h, based on the specification with CSA-by-year fixed effects, appear together with 90% confidence intervals in Fig. 3.[10] Horizon $h = 0$ denotes the investment year; $h = 1$ corresponds to the one-year lagged coefficient reported in Table 2. Payroll responds immediately and significantly at all horizons, consistent with rapid wage adjustments as VC-backed firms compete for talent. Employment rises significantly from $h = 1$, and new ventures—requiring physical setup—respond most clearly from $h = 3$. This sequencing aligns with the expectation that wage adjustments precede hiring, which in turn precedes new venture formation.

5.2. IV results

Our rich set of fixed effects cannot fully absorb unobserved shocks at the county-year level that may simultaneously attract VC investment and drive local economic performance. To address endogeneity we turn to instrumental variable estimation. Table 3 reports the results of instrumenting inward VC investment with SAVC, the social access of a county to the wider venture capital investment cycle. SAVC varies over time as VC firms adjust investment activity, and across space, both according to local differences in venture capital investment cycles in source counties, and the strength of social ties between source and destination counties. By excluding all flows into the destination county's own MSA, SAVC is designed to capture non-local supply-side variation. In addition, we control for a set of percentile dummies of SAGDP, social-access-weighted GDP of other counties, thereby absorbing general market-access effects. Because county-level GDP is available only from 2001, the sample size is reduced accordingly.

Panel A of Table 3 presents the second stage and Panel B the first stage results. Starting with Panel B, SAVC significantly predicts inward VC investment: a one-standard-deviation increase in SAVC (one unit) raises investment by 7.4% and cluster-robust Kleibergen–Paap F-statistics exceed 10 in all specifications, indicating instrument strength (Stock and Yogo, 2005). Difference-in-Sargan χ^2 tests reject exogeneity of VC investment in the OLS specification ($p < 0.001$ in all columns).[11]

[10] We expand Eq. (2) as follows: $Y_{i,t+h} = \beta_h VC_{i,t} + \gamma' X_{i,t} + \mu_i + \nu_{rt} + \varepsilon_{i,t}$. A so-called lag-augmented version with in addition one lag of both Y and VC included on the right-hand-side results in a very similar pattern.

[11] Using Stata's ivreg2, endog() option (Baum et al., 2010).

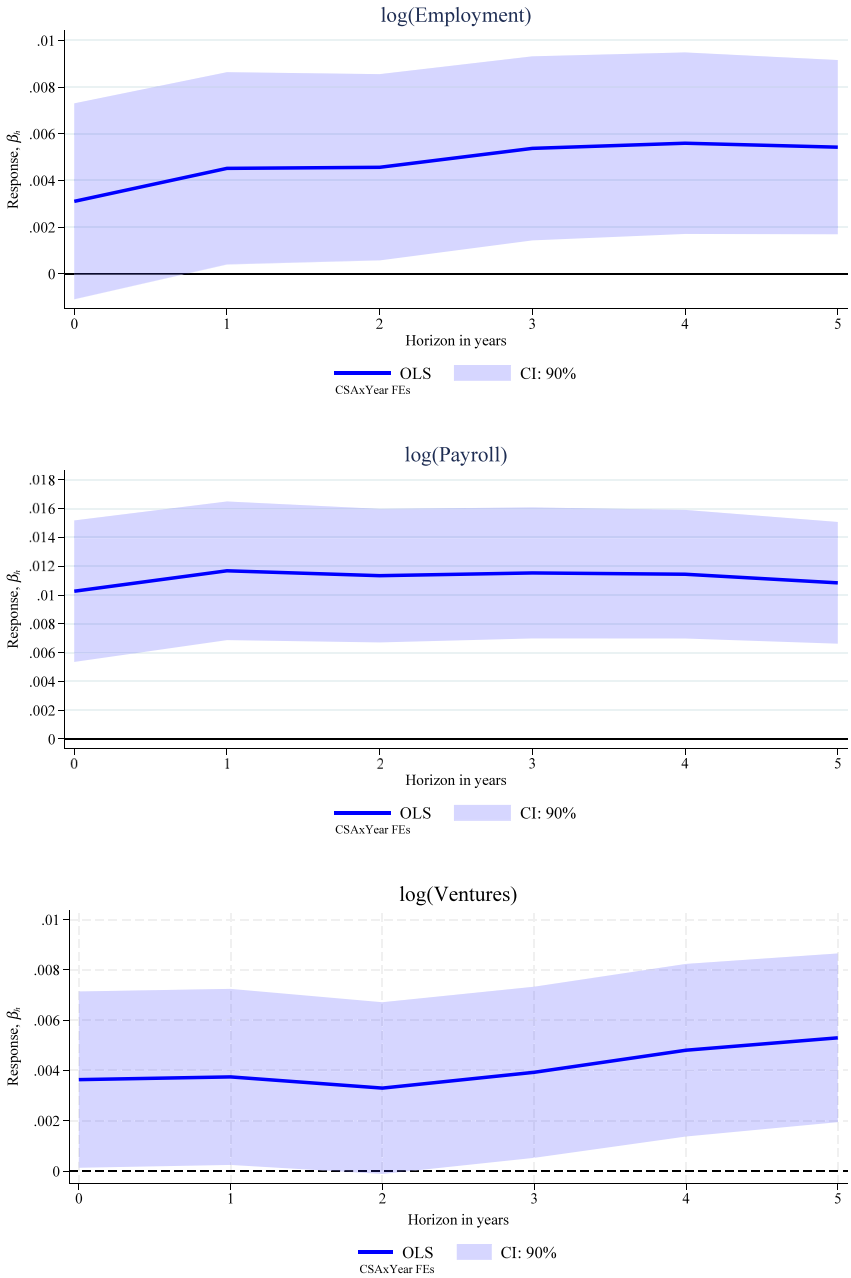

Fig. 3. Effect of Inward VC investments over time (local projections method of Jorda and Taylor, 2025).

Panel A shows that the IV estimates are substantially larger than their OLS counterparts: doubling VC investment increases employment by at least 11%, payroll by at least 12%, and new ventures by at least 11%. These county-level estimates capture both direct effects on VC-backed firms and spillovers to other local firms; Section 6 disentangles industry-level channels.

The IV coefficients are larger than OLS. Three factors may explain this: (i) classical measurement error in VC flows, which attenuates OLS; (ii) the IV estimand is a local average treatment effect (LATE) for counties whose inward VC investments respond to VC supply shocks and their social access to these—plausibly those with higher marginal returns; and (iii) the inverse-hyperbolic-sine transformation is not scale-invariant (Chen and Roth, 2024). We view measurement error (see also Maats et al., 2011 and Kaplan and Lerner, 2016) and LATE heterogeneity as the most likely explanations; robustness to alternative transformations (available on request) suggests that explanation (iii) is secondary.

Following Mullahy and Norton (2024), we also present IV results for *untransformed* outcomes in Table 4. This specification relaxes log-linearity and directly shows the impact of each million dollars invested on employment, dollar wages, and new ventures. The first

Table 3
IV regressions of county-level aggregate outcomes on inward VC investments.

Panel A: Second stage

Dependent variable →	log Employment		log Payroll		log Ventures	
	(1)	(2)	(3)	(4)	(5)	(6)
Inward VC Investment, asinh, t-1	0.180***	0.112***	0.217***	0.125***	0.167***	0.111***
	(0.050)	(0.036)	(0.061)	(0.041)	(0.046)	(0.030)

Panel B: First stage

Dependent variable →	Inward VC Investment, asinh					
Social Access to VC (SAVC)	0.074***	0.083***	0.074***	0.083***	0.074***	0.083***
	(0.016)	(0.016)	(0.016)	(0.016)	(0.016)	(0.016)
Observations	55,854	55,872	55,986	56,004	56,230	56,248
First-stage F-stat	20.8	27.0	20.9	27.0	20.9	26.9
Difference-in-Sargan test χ^2	37.2	21.2	17.9	12.9	62.9	39.6
p-val	0.000	0.000	0.000	0.000	0.000	0.000
Anderson–Rubin χ^2	26.4	12.9	28.6	11.9	30.0	22.8
p-val	0.000	0.000	0.000	0.001	0.000	0.000
County FEs	Yes	Yes	Yes	Yes	Yes	Yes
State×Year FEs	Yes	No	Yes	No	Yes	No
CSA×Year FEs	No	Yes	No	Yes	No	Yes
SAGDP FEs (100 bins)	Yes	Yes	Yes	Yes	Yes	Yes

Note: Standard errors clustered at the county level in parentheses. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$. *Employment* is the total of full- and part-time employees on the venture's payroll; *Payroll* includes all forms of compensation paid during the reporting year in thousands of dollars; *Ventures* is the count of individual physical locations at which business is conducted. *Inward VC Investment* is the inverse hyperbolic sine (asinh) of all inward VC equity investments in millions of dollars. CSA×Year FEs are combined statistical area-by-year fixed effects. SAGDP FEs are 100 fixed effects, one for each percentile of SAGDP. The first-stage F-stat is the cluster robust Kleibergen–Paap Wald statistic. *Difference-in-Sargan test* χ^2 reports the robust difference-in-Sargan test of endogeneity of the endogenous regressor (using the endog option in Stata's ivreg2), and *p-val* reports the *p*-value for this test. *Anderson–Rubin* χ^2 reports the Anderson–Rubin test of significance of the endogenous regressor, and *p-val* reports the *p*-value for this test.

stage is weaker: Kleibergen–Paap F-statistics fall below 10, likely because the instrument struggles to predict the largest investments (which log transformation would down-weight). Despite this, the weak-instrument robust Anderson–Rubin χ^2 test of the second stage coefficient obtains better significance levels than the second-stage t-stats and also clearly rejects the null that VC has no effect. We nevertheless address this issue more thoroughly by following Andrews et al. (2019): we report Montiel and Pflueger (2013) effective F-statistics—which correct for non-homoskedastic errors—in Panel B, and weak identification-robust and efficient Anderson–Rubin (AR) confidence intervals are reported in Panel A. The AR intervals yield unbounded upper limits but provide robust *lower* bounds. Each million dollars invested yields at least one new venture, 41 jobs, and $7 million in payroll (implying relatively high-paying jobs). This implies a multiplier of roughly 7. For comparison, while fiscal multipliers range from 0.6 to 1.5 in normal times (Ramey, 2016), and rise to 2–5 under accommodative monetary policy (Cloyne et al., 2020; Fernández Romero, 2025) or for public R&D spending (Antolin-Diaz and Surico, 2025), with recent estimates for innovation-oriented expenditure approaching 10 (Ciaffi et al., 2024).

5.3. Robustness

The main results in Sections 5.1 and 5.2 prove robust to several alternative specifications and variable definitions. We organize these tests into three categories and summarize key findings here; full tables appear in the Online Appendix. All robustness regressions augment the preferred IV specification (Table 3, columns 2, 4, 6 that include CSA-by-year fixed effects).

Instrument definitions. The baseline SAVC instrument (Eq. (3)) is defined as the SCI-weighted sum of all (outgoing) VC investments from all source countries, but excludes any investments flowing into the focal county's MSA. In robustness checks, we vary SAVC by (i) excluding only the destination county itself, (ii) excluding the destination county's CSA, (iii) excluding source counties in the destination's MSA, and (iv) combining (ii) and (iii). Formulas are provided in Online Appendix OA2. Table OA3 shows that results are stable across definitions; excluding own-MSA origins (definitions iii and iv) yields larger point estimates, as expected when removing the potentially most endogenous flows.

Clustering. In the main analysis we always cluster standard errors by county, from the logic that this is also the level of treatment. In Online Appendix OA3 we cluster standard errors alternatively by state, CSA, or two-way (county and year). Standard errors widen modestly under state clustering, but all coefficients and F-statistics remain significant.

Table 4
IV regressions of untransformed variables.

Panel A: Second stage

Dependent variable →	Employment		Payroll		Ventures	
	(1)	(2)	(3)	(4)	(5)	(6)
Inward VC Investment, t-1	78.0**	83.8**	16,916.6**	17,829.7**	2.3*	3.0**
	(37.2)	(37.6)	(8471.7)	(8509.7)	(1.2)	(1.5)
Anderson–Rubin 95% C.I.	[41.5, ...]	[47.7, ...]	[7267.5, ...]	[8990.1, ...]	[1.1, ...]	[1.6, ...]

Panel B: First stage

Dependent variable →	Inward VC Investment					
Social Access to VC (SAVC)	73.8*	62.4**	73.8*	62.4**	73.8*	62.3**
	(38.3)	(30.8)	(38.3)	(30.8)	(38.3)	(30.8)
Observations	55,872	55,872	56,004	56,004	56,248	56,248
First-stage Effective F-stat	3.7	4.1	3.7	4.1	3.7	4.1
Difference-in-Sargan test χ^2	9.6	13.0	2.3	3.5	4.0	13.4
p-val	0.002	0.000	0.133	0.061	0.046	0.000
Anderson–Rubin χ^2	21.7	24.7	12.5	16.3	17.2	19.9
p-val	0.000	0.000	0.000	0.000	0.000	0.000
County FEs	Yes	Yes	Yes	Yes	Yes	Yes
State×Year FEs	Yes	No	Yes	No	Yes	No
CSA×Year FEs	No	Yes	No	Yes	No	Yes
SAGDP FEs (100 bins)	Yes	Yes	Yes	Yes	Yes	Yes

Note: Standard errors clustered at the county level in parentheses. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$. *Employment* is the total of full- and part-time employees on the venture's payroll; *Payroll* includes all forms of compensation paid during the reporting year in thousands of dollars; *Ventures* is the count of individual physical locations at which business is conducted. *Inward VC Investment* is all inward VC equity investments in millions of dollars. County×Five Year Period FEs are county-by-period fixed effects where periods are five-year long non-overlapping periods starting with 1986–1990. CSA×Year FEs are combined statistical area-by-year fixed effects. SAGDP FEs are 100 fixed effects, one for each percentile of SAGDP. The first-stage effective F-stat is the Montiel and Pflueger (2013) statistic. Anderson–Rubin 95% C.I. are coefficient confidence intervals that are robust to potential weak identification and are efficient in the one-instrument case (Andrews et al., 2019). When the algorithm did not find an (upper) bound we report "...". *Difference-in-Sargan test* χ^2 reports the robust difference-in-Sargan test of endogeneity of the endogenous regressor (using the endog option in Stata's ivreg2), and *p-val* reports the *p*-value for this test. *Anderson–Rubin* χ^2 reports the Anderson–Rubin test of significance of the endogenous regressor, and *p-val* reports the *p*-value for this test.

Exclusion-restriction controls. Online Appendix OA4 varies the SAGDP control from 50 to 500 bins, a linear term, or omission altogether. Results are stable; omitting SAGDP raises coefficients by 20%–40%, underscoring the importance of controlling for general market access.

Excluding california, massachusetts, and new york. To ensure results are not driven by the largest VC hubs, Online Appendix OA5 drops California, Massachusetts, and/or New York from the sample (see also Fig. 1). Point estimates remain significant and, if anything, increase when California is excluded.

Human-capital controls. A remaining exclusion-restriction concern is that socially connected counties may also be skill-abundant and systematic different in terms of investment opportunities. For example, cities like San Francisco may also be particularly active in patenting, startup activity, host major universities, and have a high concentration of skilled labor (Bloom et al., 2025). Online Appendix OA6 adds the county-year share of adults with a bachelor's degree. The coefficient drops 30%–50%, but this decline tracks the smaller sample (counties with at least 65,000 population, observed in 2005–2019); within comparable samples, adding education leaves estimates virtually unchanged.

5.4. Entry and exit

The preceding estimates capture net county-level changes. Underlying these aggregates is churn: companies entering and exiting the market, jobs being created and destroyed. H2b predicts that entry and job creation outweigh exit and job destruction when VC investment succeeds at scale. Table 5 reports OLS regressions with the dependent variable set to the log number of ventures entering or exiting. Job creation is defined as the log of total jobs added at new and growing ventures, and job destruction is defined analogously for exiting and shrinking ventures. Data come from the US Census Business Dynamics Statistics. Across both specifications, VC investment is positively associated with venture entry and job creation; effects on exit and job destruction are smaller and less robust. Instrumenting VC investment in Table 6 shows a broadly similar pattern: entry and job creation effects

Table 5
OLS Regressions of county-level business dynamics outcomes on VC investments.

Dependent variable →	log Venture Entry		log Venture Exit		log Job Creation		log Job Destruction	
	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)
Inward VC Investment, asinh, t-1	0.015***	0.008***	0.010***	0.003	0.018***	0.011***	0.010***	0.002
	(0.002)	(0.002)	(0.002)	(0.002)	(0.002)	(0.002)	(0.002)	(0.003)
Observations	100,972	101,005	101,037	101,070	102,749	102,782	102,746	102,779
Adjusted R^2	0.976	0.974	0.974	0.972	0.969	0.968	0.966	0.964
County FEs	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
State×Year FEs	Yes	No	Yes	No	Yes	No	Yes	No
CSA×Year FEs	No	Yes	No	Yes	No	Yes	No	Yes

Note: Standard errors clustered at the county level in parentheses. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$. *Venture Entry (Exit)* is the number of ventures entering (exiting) the market. *Job Creation (Destruction)* is the number of jobs added at entering and growing (exiting and shrinking) ventures. *Inward VC Investment* is the inverse hyperbolic sine (asinh) of all inward VC equity investments in millions of dollars. CSA×Year FEs are combined statistical area-by-year fixed effects.

Table 6
IV Regressions of county-level business dynamics outcomes on VC investments.

Panel A: Second stage

Dependent variable →	log Venture Entry		log Venture Exit		log Job Creation		log Job Destruction	
	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)
Inward VC Investment, asinh, t-1	0.199***	0.159***	0.140***	0.119***	0.309***	0.198***	0.076	−0.026
	(0.059)	(0.045)	(0.040)	(0.035)	(0.079)	(0.054)	(0.051)	(0.047)

Panel B: First stage

Dependent variable →	Inward VC Investment, asinh							
Social Access to VC (SAVC)	0.073***	0.082***	0.073***	0.082***	0.073***	0.082***	0.073***	0.082***
	(0.016)	(0.016)	(0.016)	(0.016)	(0.016)	(0.016)	(0.016)	(0.016)
Observations	54,943	54,961	55,030	55,048	56,050	56,068	56,044	56,062
First-stage F-stat	19.9	25.0	19.8	25.0	20.0	25.0	20.0	25.0
Difference-in-Sargan test χ^2	24.5	22.7	16.4	12.7	33.2	28.5	2.6	0.3
p-val	0.000	0.000	0.000	0.000	0.000	0.000	0.107	0.602
Anderson–Rubin $\chi 2$	17.4	19.7	17.2	16.7	34.7	21.5	2.2	0.4
p-val	0.000	0.000	0.000	0.000	0.000	0.000	0.140	0.554
County FEs	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
State×Year FEs	Yes	No	Yes	No	Yes	No	Yes	No
CSA×Year FEs	No	Yes	No	Yes	No	Yes	No	Yes
SAGDP FEs (100 bins)	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes

Note: Standard errors clustered at the county level in parentheses. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$. *Venture Entry (Exit)* is the number of ventures entering (exiting) the market. *Job Creation (Destruction)* is the number of jobs added at entering and growing (exiting and shrinking) ventures. *Inward VC Investment* is the inverse hyperbolic sine (asinh) of all inward VC equity investments in millions of dollars. CSA×Year FEs are combined statistical area-by-year fixed effects. SAGDP FEs are 100 fixed effects, one for each percentile of SAGDP. The first-stage F-stat is the cluster robust Kleibergen–Paap Wald statistic. *Difference-in-Sargan test* χ^2 reports the robust difference-in-Sargan test of endogeneity of the endogenous regressor (using the endog option in Stata's ivreg2), and *p-val* reports the *p*-value for this test. *Anderson–Rubin* χ^2 reports the Anderson–Rubin test of significance of the endogenous regressor, and *p-val* reports the *p*-value for this test.

strengthen, and exit rises modestly but remains dominated by entry. Local projections (Online Appendix OA7, Figure OA4) confirm that entry persists over five years, job creation peaks early, and exit/destruction remain muted through year 3.

In sum, VC investment generates firm entry and job creation at the county level, with limited evidence of offsetting exit or job destruction. This pattern suggests that (i) local competition effects are modest, or (ii) they are likely offset by innovation spillovers—consistent with H2b.

Table 7
County-by-industry level OLS regressions of employment on VC investments.

Dependent variable →	log Employment			
	(1)	(2)	(3)	(4)
Inward VC Investment in Same Industry, asinh, t-1	0.111***	0.084***	0.143***	0.107***
	(0.007)	(0.007)	(0.008)	(0.008)
Inward VC Investment in Other Industries, asinh, t-1			0.112***	0.079***
			(0.014)	(0.013)
Observations	1,712,264	1,711,176	1,712,264	1,711,176
Adjusted R^2	0.779	0.796	0.779	0.796
County×Year FEs	Yes	Yes	Yes	Yes
Industry×Year FEs	Yes	No	Yes	No
Industry×CSA×Year FEs	No	Yes	No	Yes

Note: Standard errors clustered at the county in parentheses. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$. *log Employment* is the (LN) total of full- and part-time employees on the venture's payroll aggregated by county-industry-year. *Inward VC Investment in Same Industry* is the inverse hyperbolic sine (asinh) of all inward VC equity investments in millions of dollars in the same industry as the dependent observation; *Inward VC Investment in Other Industries* is the inverse hyperbolic sine (asinh) of all inward VC equity investments in millions of dollars in industries other than the dependent observation.

6. Industry spillovers

The preceding county-level estimates capture the combined effect of VC on all local industries. This section disaggregates those effects to test whether VC investment spills over to industries that do not themselves receive funding (H3) and whether own-industry effects remain positive net of local competition (H2b). If spillovers are substantial, VC contributes to the broader entrepreneurial ecosystem—not only by scaling funded ventures but also by expanding the local ecosystem of suppliers that supports future entrepreneurship.

We disaggregate to the industry–county–year level using the 22 top-level NAICS (2012) industries. We distinguish two channels: (1) the effect on an industry's employment when that *own* industry receives VC (testing H2b), and (2) the effect when *other* industries in the same county receive VC (testing H3). Own-industry VC captures the most direct effects—scaling, entry, and intra-industry competition—while other-industry VC captures spillovers through input–output linkages. We estimate Eq. (5) by OLS, relying on more detailed county-by-year and industry-by-CSA-by-year fixed effects for identification.[12] Table 7 presents the results.

Columns 1 and 2 restrict attention to own-industry VC (thus assuming $\delta_2 = 0$). A doubling of own-industry VC investment is associated with roughly a 10% increase in industry employment—consistent with the county-level magnitudes in Section 5. Columns 3 and 4 add other-industry VC. The coefficient on other-industry VC is positive and significant: a doubling is associated with a 7%–8% employment increase. Comparing coefficients within the same specification, own-industry VC has an effect roughly 25% larger in absolute terms than other-industry VC, and F-tests reject equality at the 5% level ($p = 0.017$ and $p = 0.024$, respectively, for columns 3 and 4). These results support H3: VC investment generates positive spillovers to industries that do not receive funding directly, plausibly through upstream supplier linkages.

Fig. 4 traces these effects over a five-year horizon using local projections (Jorda and Taylor, 2025), based on the specification in column 4 of Table 7. Both own-industry and other-industry effects are statistically significant at all horizons. Notably, the gap between the two coefficients narrows over time: the p-value for an F-test of equality rises from 0.02 at horizon 1 to 0.06 at horizon 5 (right axis). This pattern is consistent with the idea that inter-industry spillovers accumulate gradually—likely as upstream suppliers expand capacity in response to sustained demand from VC-backed sectors. The growing spillover channel helps explain the persistent county-level employment gains documented in Section 5.

Which industries benefit most? Online Appendix OA9 disaggregates these results further by two-digit NAICS industry. The three largest VC recipients are 'Information', 'Manufacturing', and 'Professional Services'. Yet the industries with the largest employment responses to county-level VC are business-services sectors—particularly 'Management of Companies and Enterprises' (NAICS 55) and 'Professional, Scientific, and Technical Services' (NAICS 54). Strikingly, these sectors benefit primarily from *other-industry* VC rather than own-industry investment, exhibiting the largest spillover coefficients and small or zero own-industry effects. Information ranks fourth, while Manufacturing shows relatively muted employment gains, perhaps because VC in that sector emphasizes capital investment and labor-saving technology. We also observe negative other-industry coefficients for some sectors (e.g., Retail Trade, Manufacturing), indicating labor reallocation away from industries that compete for workers. Taken together, these patterns suggest that a substantial share of VC's employment impact operates through upstream supplier linkages, reinforcing the resource munificence of local entrepreneurial ecosystems (Stam and van de Ven, 2021).

[12] IV would require two instruments, splitting (network distance to) total outward VC sources by same/other industry. Exogenous variation between the two would however be limited because other- and same-industry VC supply from the same source would be influenced by correlated shocks, making it harder to separately identify same- and other-industry VC inflow. Unreported tests suggest that the instruments indeed suffer from severe weakness when used for two endogenous variables. However, the Difference-in-Sargan χ^2 test rejects in all cases that the endogenous variables should be treated as endogenous, supporting the use of OLS combined with highly granular fixed effects.

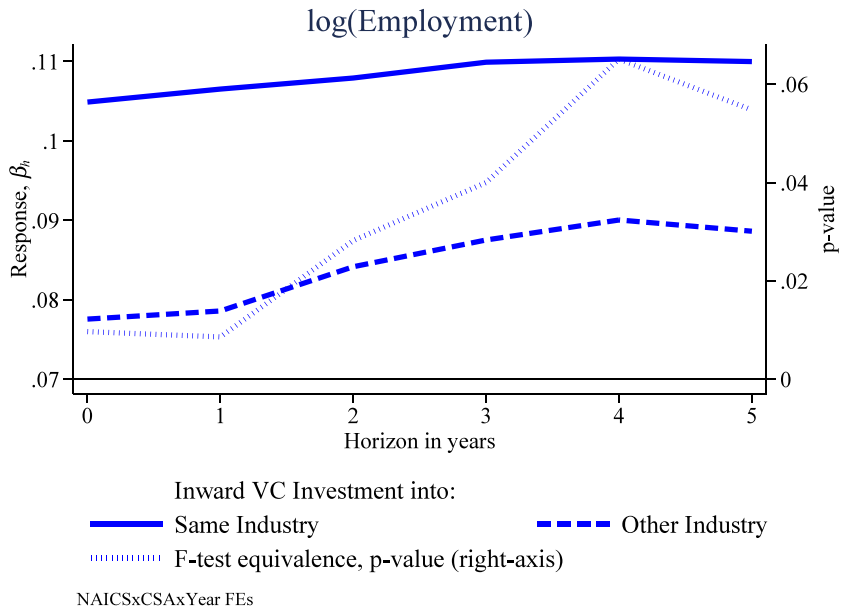

Fig. 4. Effect of Inward VC investments over time for same or other industries (local projections method of Jorda and Taylor, 2025).

Job creation and destruction by county-industry. To complement the aggregate results on job creation and destruction in Section 5.4, we examine job dynamics within county-industry cells. This allows us to assess whether the net employment gains documented above are partially offset by job destruction in some firms, and whether reallocation occurs across industries (H2b and H3). We estimate variants of Eq. (5) with job creation and job destruction as outcomes, using data provided by the Business Dynamics Statistics (BDS).[13]

Results appear in Online Appendix OA8. Job creation (Table OA8) rises strongly with own-industry VC: the coefficient is positive and highly significant across all specifications. Job creation also responds positively to other-industry VC, consistent with upstream and business-services spillovers (H3). Job destruction also rises with own-industry VC (Table OA9), consistent with creative destruction and labor reallocation toward more productive firms. However, the coefficient on job creation exceeds that on destruction in comparable specifications, implying net positive employment growth within treated industries—supporting H2b.

Local projections (Figure OA5) confirm that job creation responses to own-industry VC persist over five years, while destruction responses are somewhat smaller and also sustained. The response of both outcomes to other-industry VC is weaker but rises after year 3, consistent with the gradual emergence of supplier-side spillovers.

In sum, the county-industry evidence supports two mechanisms. First, own-industry VC generates net job creation despite elevated destruction, consistent with H2b: the scaling of funded ventures and knowledge spillovers outweigh displacement of less productive incumbents. Second, other-industry VC is associated with net job creation and increased labor reallocation (since both job creation and destruction go up) in supplier sectors, consistent with H3 and the broader ecosystem-development channel.

7. Discussion and conclusions

This paper asks whether venture capital investment generates economic activity beyond the funded ventures themselves. Using a novel instrument—Social Access to Venture Capital (SAVC)—and detailed data on investor-to-venture flows across U.S. counties from 1986 to 2019, we find that it does: VC investment increases county-level employment, payroll, and venture creation, and these effects extend to industries that do not themselves receive VC funding. These benefit indirectly through knowledge spillovers and business-to-business buyer–supplier relations and contribute to the local entrepreneurial ecosystem.

Summary of findings. Our conservative IV estimates indicate that each million dollars of VC investment produces at least one new venture, 41 jobs, and $7 million in payroll. In elasticity terms, a doubling of VC investment yields at least an 11% increase in ventures, an 8% increase in employment, and a 10% increase in payroll. These results are robust to alternative instrument definitions, rich fixed-effects specifications, clustering choices, and the exclusion of California, Massachusetts, and New York.

[13] We focus on job creation and destruction in this section, because, although the US Census Bureau provides data on firm entry and firm exit at the county-industry-year level, these data have around 45% of county-industry-year observations missing or suppressed. In comparison, the rate of missing for job creation and destruction is a lot lower, around 22% of observations.

Linking these results to our theoretical framework, the positive net effects on county-level employment and venture creation support H1: successful venture scaling outweighs portfolio failure rates. This finding is also consistent with the view that VC helps build successful companies rather than merely selecting firms that would grow regardless. Payroll increases confirm H2a, reflecting labor-market competition for talent and productivity gains. The dominance of venture entry and job creation over exit and destruction supports H2b, suggesting that knowledge spillovers and new entry outweigh within-industry displacement.

Our most distinctive finding concerns inter-industry spillovers (H3). Several industries that do not themselves receive VC, particularly Management of Companies and Enterprises (NAICS 55) and Professional, Scientific, and Technical Services (NAICS 54), experience significant employment gains when *other* local industries attract VC funding. These upstream supplier sectors expand as a critical resource—alongside physical and financial inputs, human capital, and demand—reinforcing the local entrepreneurial ecosystem and reducing entry barriers for future entrepreneurs (Stam and van de Ven, 2021). Notably, inter-industry effects start smaller than within-industry effects but converge over a five-year horizon, suggesting that the supplier base develops gradually in response to sustained demand from VC-backed sectors. We also observe labor reallocation: some industries shed employment as resources shift toward VC-intensive sectors. Yet this reallocation may benefit workers on net, as it channels labor toward higher-productivity, higher-paying firms.

Theoretical implications. Our findings advance knowledge of venture capital's role in local economic development in three ways. First, our local focus helps unpack the more aggregate country- or industry-wide estimates of VC's impact (e.g., Akcigit et al., 2022; Popov and Roosenboom, 2012; Popov, 2014), bringing the analysis closer to the level at which entrepreneurial ecosystems operate. We update and extend Samila and Sorenson (2011) to a period (1986–2019) characterized by the geographic decentralization of VC activity and by industry-level analysis across all U.S. counties. Our IV estimates are larger than those reported for the 1993–2002 period, though direct comparison is complicated by differences in identification strategy and sample composition; the LATE identified by our SAVC instrument may apply to counties with particularly high marginal returns to VC.

Second, the novel evidence of inter-industry spillovers to upstream supplier sectors highlights an understudied channel through which VC contributes to entrepreneurial ecosystems. Most research on ecosystem emergence has emphasized top-down mechanisms such as facilitating individual interactions, cultural roots, incubator selection, university knowledge spillovers, and local policy interventions (such as seeding actions and creating opportunities and support) (Huggins et al., 2024; Bergman and McMullen, 2026; Juma et al., 2023; Thompson et al., 2018; Mack and Mayer, 2016). Our results point to a complementary, demand-driven channel: VC-backed ventures generate demand for business-to-business services, which expands the local supplier base and thereby enhances the resource endowment available to subsequent entrepreneurs (Stuart and Sorenson, 2003; Stam, 2015). The implicit assumption in much of the ecosystem literature, that new ventures are largely self-sufficient in their day-to-day operations beyond funding, may understate the importance of locally available producer services. These may support the endogenous emergence of ecosystems.

Third, our results suggest that the decline in corporate governance during the 2010s (Lerner and Nanda, 2020) and the reduction in entrepreneurial quality observed during the 2000s (Guzman and Stern, 2020) did not prevent VC from generating positive aggregate local effects, possibly because federal support for entrepreneurial clusters provided countervailing support (Chatterji et al., 2014) or because these trends proved temporary.

Practical implications. What can local policymakers do to attract VC investment? Our finding that social connectedness to VC source counties predicts inward investment points to policies that strengthen ties between local entrepreneurs and non-local (and out of state) investors. Beyond place-based incentives targeted directly at investors or startups, investments in transportation infrastructure that reduce travel time to VC hubs may indirectly stimulate VC activity by facilitating face-to-face interactions and repeat engagement (Bernstein et al., 2016; Hornbeck and Rotemberg, 2024). Counties may also cultivate business-related social ties through initiatives that promote repeated interaction, such as trade shows, startup competitions, accelerator and incubator programs, and industry conferences, possibly orchestrated by 'ecosystem architects' (Bergman and McMullen, 2026; Crisanti et al., 2021). The inter-industry spillover evidence further suggests that policies facilitating business-to-business linkages can expand the scope of the local entrepreneurial ecosystem to include suppliers from neighboring locations.

Limitations and future research. Several limitations warrant discussion. First, the CBP count of establishments includes both de novo startups and new locations opened by multi-establishment firms; our data do not distinguish between the two, which would require detailed ownership records. Second, although our data cover all U.S. counties over three decades, we do not observe the firm-to-firm linkages that theory emphasizes; the inter-industry spillovers we document are inferred from industry-level aggregates rather than traced through specific buyer–seller relationships. Third, because we present both OLS results relying on granular fixed effects and IV lower-bound estimates from 2SLS regressions, the magnitudes may be sensitive to the identification method. With controlled experiments being infeasible, causal evidence remains challenging in a setting that covers many outcomes, locations, industries, and years.

These limitations suggest several directions for future research. Matched employer–employee data, buyer–supplier networks, or patent-citation flows could trace how VC-induced growth propagates through specific channels. Future research may also identify the spatial boundaries and reach of both existing and nascent ecosystems and examine how they develop over time, accounting for linkages to other regions. Future work may revisit the trends in corporate governance and entrepreneurial quality with more recent data, and could benefit from alternative natural experiments, such as exogenous policy changes or spatially differentiated funding shocks, that complement the SAVC instrument we propose here. Finally, our county-level focus captures localized effects; understanding cross-county spatial spillovers through commuting, trade, and migration remains an important open question.

CRediT authorship contribution statement

Steven Poelhekke: Writing – review & editing, Writing – original draft, Validation, Supervision, Resources, Project administration, Methodology, Investigation, Conceptualization. **Benjamin Wache:** Writing – review & editing, Writing – original draft, Visualization, Methodology, Formal analysis, Data curation, Conceptualization.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Appendix A. Supplementary data

Supplementary material related to this article can be found online at https://doi.org/10.1016/j.jbusvent.2026.106588.

Data availability

Data will be made available on request.

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